

vtable package for summary tables

Load packages

```
library(tidyverse)
library(kableExtra)
library(vtable)
knitr::opts_chunk$set(echo = T)
```

Create summary table

```
# select variables to be included in summary table and provide meaningful names
df <- mtcars %>%
  select(`Miles per gallon` = mpg, `Weight (lb/1000)` = wt, `Engine size (cub in)` = disp,
         `1/4 mile time (sec)` = qsec, Horsepower = hp, am)

# create summary table
sum_tab <- st(select(df, -am), out = 'return', add.median = T, digits = 1, fixed.digits = T)

# pipe to kable
sum_tab %>%
  kable(align = 'r', booktabs = T, caption = 'Summary statistics') %>%
  kable_styling(latex_options = 'HOLD_position')
```

Table 1: Summary statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 50	Pctl. 75	Max
Miles per gallon	32	20.1	6.0	10.4	15.4	19.2	22.8	33.9
Weight (lb/1000)	32	3.2	1.0	1.5	2.6	3.3	3.6	5.4
Engine size (cub in)	32	230.7	123.9	71.1	120.8	196.3	326.0	472.0
1/4 mile time (sec)	32	17.8	1.8	14.5	16.9	17.7	18.9	22.9
Horsepower	32	146.7	68.6	52.0	96.5	123.0	180.0	335.0

Create conditional summary table

```
# create summary table for automatic transmission vehicles only (am == 0)
automatic <- st(filter(df, am == 0), out = 'return', add.median = T, digits = 1, fixed.digits = T) %>%
  mutate(Transmission = 'Automatic') %>%
  filter(Variable != 'am')

# create summary table for manual transmission vehicles only (am == 1)
manual <- st(filter(df, am == 1), out = 'return', add.median = T, digits = 1, fixed.digits = T) %>%
  mutate(Transmission = 'Manual') %>%
  filter(Variable != 'am')

# vertically combine tables
cond_sum_tab <- rbind(automatic, manual) %>%
  relocate(Transmission)

# pipe to kable
cond_sum_tab %>%
  kable(align = 'lrrrrrrrrr', booktabs = T, caption = 'Conditional summary statistics') %>%
  collapse_rows(columns = 1, latex_hline = "major") %>%
  kable_styling(latex_options = "HOLD_position")
```

Table 2: Conditional summary statistics

Transmission	Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 50	Pctl. 75	Max
Automatic	Miles per gallon	19	17.1	3.8	10.4	14.9	17.3	19.2	24.4
	Weight (lb/1000)	19	3.8	0.8	2.5	3.4	3.5	3.8	5.4
	Engine size (cub in)	19	290.4	110.2	120.1	196.3	275.8	360.0	472.0
	1/4 mile time (sec)	19	18.2	1.8	15.4	17.2	17.8	19.2	22.9
	Horsepower	19	160.3	53.9	62.0	116.5	175.0	192.5	245.0
Manual	Miles per gallon	13	24.4	6.2	15.0	21.0	22.8	30.4	33.9
	Weight (lb/1000)	13	2.4	0.6	1.5	1.9	2.3	2.8	3.6
	Engine size (cub in)	13	143.5	87.2	71.1	79.0	120.3	160.0	351.0
	1/4 mile time (sec)	13	17.4	1.8	14.5	16.5	17.0	18.6	19.9
	Horsepower	13	126.8	84.1	52.0	66.0	109.0	113.0	335.0